

DIRECT-TO-MOBILE TECH: Solution For Poor Connectivity Or Lack Of Internet Access

D2M technology holds transformative potential but demands collaboration, innovation, and bold action for adoption. Could it redefine content consumption?



DR DEEPAK HALAN
is Professor with
Jaipuria Institute of
Management, NOIDA.

The exponential growth in mobile devices and internet connectivity over the last decade has transformed everyday life. Smartphones have become essential, particularly for accessing video and multimedia content. However, interruptions in internet connectivity, like when streaming your favourite TV show, remain a common frustration.

Innovative solutions that exploit broadcasting networks to deliver video, digital, and multimedia content directly to mobile devices without relying on internet connectivity are essential. Although 5G technology promises a significant increase in network capacity and consistent high speed, covering India's vast population will take several years.

Many previous efforts have faltered due to challenges such as the high capital costs of infrastructure, the need for specialised hardware and software to ensure device compatibility, and regional standards requiring manufacturers to develop multiple models.

"The technology is already mature and available. Some lab trials and field trials have been done. Now we must do a citywide pilot trial so that its potential can be demonstrated. The government, as a regulator, then must enable its rollout and commercial deployment. I'm hoping in the next year or so, we can launch this technology with an indigenously developed system."

—**Abhay Karandikar, Science and Technology Secretary (Source: Business Standard)**

These factors hindered cost efficiency and scalability.

Direct-to-mobile (D2M) technology presents a transformative solution for users without reliable internet access. Functioning similarly to FM radio or direct-to-home (DTH) services, D2M offers a streamlined approach to mobile content consumption.

D2M technology is tailored to deliver a wide range of content, optimised for various screen sizes, resolutions, and capabilities. It supports interactive advertising, data casting, and supplementary content while offering key benefits such as buffer-free streaming, fewer call drops, and improved data speeds.

The technology enables telecom operators to offload bandwidth-heavy applications—such as live event streaming and over-the-top (OTT) services—from broadband networks to broadcast networks, effectively decongesting broadband infrastructure. D2M also provides flexibility and variety through real-time, on-demand, seamless access.

D2M technology: Features and functionalities

D2M technology is characterised by the following main features and functionalities:



*D2M technology is a solution when there is no or poor internet connectivity
(Source: <https://www.youtube.com>)*